DATA SCIENCE & ARTIFICIAL INTELLIGENCE & CS/IT



Calculus and Optimization

Integration

Lecture No. 01



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Recap of previous lecture



Topic

Maxima - Minima of Single Variable

Topics to be Covered



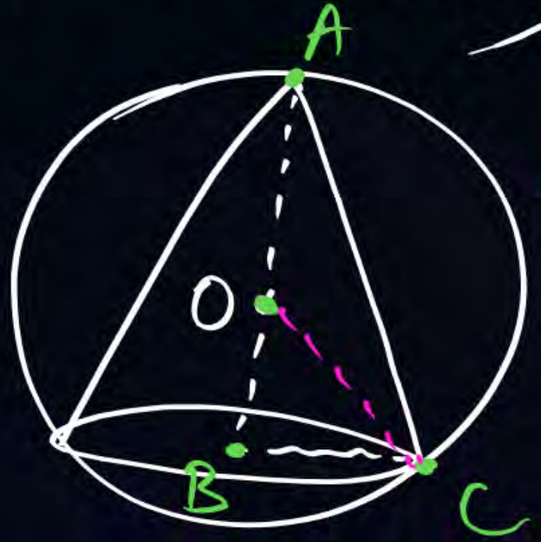
Topic

Maxima-Minima of two Variable

Qs (Type II) A Right Circular Cone of Maximum Volume is to be inscribed in a sphere of Radius 1 mtr then find the Height of such Cone.

(Application Based Questions)

Sol:



$OC = OA = 1$ mtr

Let $AB = h$, $BC = r$ } Variables

In $\triangle OBC$:

$OC^2 = OB^2 + BC^2$

$1^2 = (h-1)^2 + r^2$

$V = \frac{1}{3}\pi r^2 h$ { $V = f(r, h)$ } Surface

$V = \frac{1}{3}\pi\left[1-(h-1)^2\right] \cdot h$

Now $V = f(h)$ Curve

$V = \frac{\pi}{3}\left[2h-h^2\right]h = \frac{\pi}{3}\left(2h^2-h^3\right)$

For Max Volume, $\frac{dV}{dh} = 0$

$\Rightarrow h = 4/3$.

$V = \frac{\pi}{3}(2h^2 - h^3)$ — (1)

$\Rightarrow \frac{dV}{dh} = 0$

$\frac{\pi}{3} \cdot \frac{d}{dh}(2h^2 - h^3) = 0$

$\frac{\pi}{3}[4h - 3h^2] = 0$

$h(4 - 3h) = 0$

$h = 0$ & $h = \frac{4}{3}$

Not possible

$\frac{d^2V}{dh^2} = \frac{d}{dh}\left[\frac{\pi}{3}(4h - 3h^2)\right]$

$= \frac{\pi}{3}[4 - 6h] =$

$\left(\frac{d^2V}{dh^2}\right)_{h=\frac{4}{3}} = \frac{\pi}{3}\left[4 - 6\left(\frac{4}{3}\right)\right] = -ve$

for V is Max at $h = \frac{4}{3}$ Ans

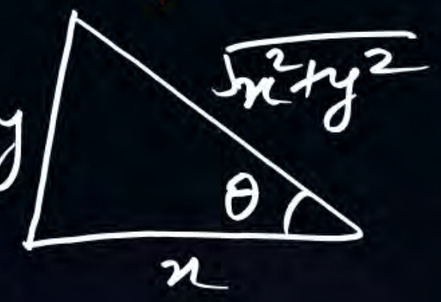
Qs In a Right angle Δ, if sum of the Hypotenuse & one side is kept constant, in order to find **Maximum area** then find the angle b/n them ?

(a) 30°

(b) 60°

(c) 45°

(d) 90°



ATQ,

$\sqrt{x^2+y^2}+x=k$

then $\theta=?$

$x^2+y^2=(k-x)^2$

$x^2+y^2=k^2+x^2-2kx$

$y^2=k^2-2kx$ — (1)

Area $(A)=\frac{1}{2}(\text{Base}\times\text{Height})=\frac{1}{2}xy$

$\Rightarrow$ Let $U=A^2=\frac{1}{4}x^2y^2$ where $A=\sqrt{U}$

$\Rightarrow A_{max}=\sqrt{U_{max}}$ = ?

$U=\frac{1}{4}x^2(k^2-2kx)$

$U=\frac{k^2x^2}{4}-\frac{k}{2}x^3$ — (2)

for Max U, Put $\frac{dU}{dx}=0$

$$\frac{d}{dx}\left[\frac{k^2x^2}{4}-\frac{k}{2}x^3\right]=0$$

$$\frac{k^2}{4}(2x)-\frac{k}{2}(3x^2)=0$$

$$\frac{k}{2}x\left[k-3x\right]=0$$

$$x\neq 0,\quad \boxed{x=\frac{k}{3}}$$

By ①, $y^2=k^2-2kx$

$$y^2=k^2-2k\left(\frac{k}{3}\right)=\frac{k^2}{3}$$

$$\boxed{y=\frac{k}{\sqrt{3}}}$$

So $\tan\theta=\dfrac{\perp^r}{Base}=\dfrac{y}{x}$

$$\tan\theta=\frac{k/\sqrt{3}}{k/3}=\sqrt{3}$$

$$\boxed{\theta=60^\circ}$$

$\sin\theta, \cos\theta, \tan\theta = \dfrac{P}{H}, \dfrac{B}{H}, \dfrac{P}{B}$

Doubt: $y^2 = k^2 - 2kx$

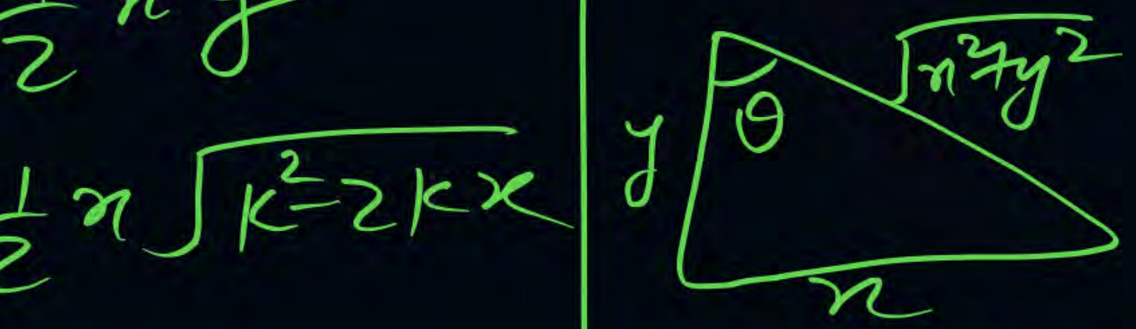
$A = \frac{1}{2} x y$

$A = \frac{1}{2} x \sqrt{k^2 - 2kx}$

$\frac{dA}{dx} = 0$

$\Rightarrow$ $x = ?$

Doubt:



$L^r = x$

Base $= y$

ATQ, $y + \sqrt{x^2+y^2} = k$

$x^2 + y^2 = (k-y)^2$

$x^2 + y^2 = k^2 + y^2 - 2ky$

$x^2 = k^2 - 2ky$

$\tan\theta = \frac{x}{y} = \sqrt{3} \Rightarrow \theta = 60^\circ$

Max-Min of func^n of two Variables (i.e of $z = f(x,y)$)



Point $\simeq (x, y)$

Value $\simeq z$

N. Cond^n for Max-Min :- $\frac{\partial z}{\partial x} = 0, \frac{\partial z}{\partial y} = 0$ — (1) — (2)

Stationary Point / Critical Point :- Points obtained by solving (1) & (2)

for eg after solving we get $x = a$ & $y = b$ then $P(a,b)$ is critical point.

Sufficient Cond^n for Max-Min :- function must satisfy Lagrange's Cond^n.

Note :- Let $P(a,b)$ is critical Point then we can use following Symbols.

$$\left(\frac{\partial^2 z}{\partial x^2}\right)_{P(a,b)} = r \quad , \left(\frac{\partial^2 z}{\partial y^2}\right)_{P(a,b)} = t \quad , \left(\frac{\partial^2 z}{\partial x \partial y}\right)_{P(a,b)} = s$$

Lagrange 's Conditions :-

① if $rt - s^2 > 0$ & $r > 0$ then $P(a,b)$ is Point of minima

② if $rt - s^2 > 0$ & $r < 0$ " " " " " MAXIMA

③ if $rt - s^2 < 0$ then " " " " Inflexion

④ if $rt - s^2 = 0$ then Case fails i.e L-Conditions are unable to give an idea about Maxima or minima & we need further investigation.

Q: The Number of Stationary Points of $Z = 2x^2 + 2xy - y^3$ is/are 2 (TWO)

(ii) Also find Optimal Values if exist?

Sol: $Z = 2x^2 + 2xy - y^3$

$Z_x = \frac{\partial Z}{\partial x} = 4x + 2y$

$Z_{xx} = \frac{\partial^2 Z}{\partial x^2} = 4$

$Z_y = \frac{\partial Z}{\partial y} = 2x - 3y^2$

$Z_{yy} = \frac{\partial^2 Z}{\partial y^2} = -6y$

$Z_{xy} = \frac{\partial^2 Z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial Z}{\partial y}\right) = \frac{\partial}{\partial x}(2x - 3y^2) = 2$

For Stationary Points, $Z_x = 0$ & $Z_y = 0$

$4x + 2y = 0$ & $2x - 3y^2 = 0$

$y = -2x$ & $2x - 3(-2x)^2 = 0$

$2x - 12x^2 = 0$

$x = 0, \frac{1}{6}$

At $x = 0, y = 0$

& at $x = \frac{1}{6}, y = -\frac{1}{3}$

i.e. $O(0,0)$ & $P\left(\frac{1}{6}, -\frac{1}{3}\right)$ are C-Points.

To check optimal value at $O(0,0)$ $\longrightarrow$

$$r = \left(\frac{\partial^2 z}{\partial x^2}\right)_O = (4)_{(0,0)} = 4$$

$$s = \left(\frac{\partial^2 z}{\partial x \partial y}\right)_O = (2)_{(0,0)} = 2$$

$$t = \left(\frac{\partial^2 z}{\partial y^2}\right)_O = (-6y)_{(0,0)} = 0$$

$\because rt - s^2 = (4)(0) - (2)^2$

$= -4$

< 0

$\Rightarrow O(0,0)$ is Saddle point.

i.e <u>No</u> optimal value exist at O

To check optimal Value at $P\left(\frac{1}{6}, -\frac{1}{3}\right)$

$$r = \left(\frac{\partial^2 z}{\partial x^2}\right)_P = (4)_P = 4$$

$$s = \left(\frac{\partial^2 z}{\partial x \partial y}\right)_P = (2)_P = 2$$

$$t = \left(\frac{\partial^2 z}{\partial y^2}\right)_P = (-6y)_P = 2$$

$\because rt - s^2 = (4)(2) - (2)^2$

$= 8 - 4 = 4 > 0$

& $r = 4 > 0$

So $P\left(\frac{1}{6}, -\frac{1}{3}\right)$ is Point of Minima

& Min Value $= [f(x,y)]_P$

$= [2x^2 + 2xy - y^3]_{P\left(\frac{1}{6}, -\frac{1}{3}\right)}$

$= \text{------} = -\frac{1}{54}$

Q: if $z = f(x,y)$ s.t $f_x(a,b) = 0$, $f_y(a,b) = 0$ — 1st Condn.

also we have,

$f_{xy}^2(a,b) - f_{xx}(a,b) f_{yy}(a,b) < 0$ & $f_{xx}(a,b) < 0$ — 2nd Condn

then $P(a,b)$ is Point of ____

(a) Minima

(b) Maxima

(c) Inflexion

(d) Data Insufficient

$\rightarrow$ $s^2 - rt < 0$ & $r < 0$

$rt - s^2 > 0$ & $r < 0$

$\therefore$ $P(a,b)$ is Point of Maxima

(2025) $z = f(x,y) = (x+y-1)^2 + (x+y)^2$ & $S = \{(x,y) : x \in R, y \in R\}$

where S contains set of points which minimizes $f(x,y)$ then

(a) S contains exactly one point

(b) S is empty

(c) S contains Infinitely many points

(d) S contains finite Number of points.

$\rightarrow$ $\text{Min } f = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}$

$z_x = 2(x+y-1) + 2(x+y)$

$z_{xx} = 2+2 = 4 = r$

$z_y = 2(x+y-1) + 2(x+y)$

$z_{yy} = 2+2 = 4 = t$

$z_{xy} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}(2x+2y-2+2x+2y) = 4 = s$

$\because rt - s^2 = (4)(4) - (4)^2 = 0$ i.e Case fails

1	3	S
2	4	?=R

(a) 4

(b) P

(c) Q

(d) ✓ R

For Critical Points:

$$\frac{\partial z}{\partial x}=0 \Rightarrow 2(x+y-1)+2(x+y)=0$$

$$\frac{\partial z}{\partial y}=0 \Rightarrow 2(x+y-1)+2(x+y)=0$$

$$2(x+y-1)+2(x+y)=0$$

or $$x+y-1+x+y=\frac{0}{2}$$

$$2x+2y=1$$

$$\boxed{x+y=\frac{1}{2}}$$

Underdetermined system $\Rightarrow \infty$ / ∞ ..... ∞.

$(x,y)=\left(\frac{1}{4},\frac{1}{4}\right),\left(\frac{3}{4},\frac{-1}{4}\right),\left(\frac{-1}{4},\frac{3}{4}\right),\left(\frac{1}{2},0\right)$

$$\frac{\partial}{\partial y}(2\sin yz) = 2\cos(yz)\cdot\frac{\partial}{\partial y}(yz)$$

$$= 2\cos(yz)\cdot(1\times z)$$

$$= 2z\cos yz$$

$w = f(x,y,z)$

THANK - YOU

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